



What a Sample Can Support

Grades 9-12 • SAT / ACT mathematics foundations

Problem-Solving and Data Analysis • Evaluate a sampling method and the limits of a claim.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Evaluate a sampling method and the limits of a claim.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



Two worked examples

Model 1. A survey of 55 people attending a poetry event finds that 36 favor a proposal. A report claims about 65% of all library members favor it. Evaluate the claim and suggest a better sampling plan.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

Model 2. A survey of 30 visitors to one sports field finds that 20 favor a proposal. A report claims about 67% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A survey of 65 volunteers from its debate club finds that 43 favor a proposal. A report claims about 66% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.

2. A survey of 65 people attending a poetry event finds that 43 favor a proposal. A report claims about 66% of all library members favor it. Evaluate the claim and suggest a better sampling plan.

3. A survey of 50 visitors to one sports field finds that 33 favor a proposal. A report claims about 66% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A survey of 55 volunteers from its debate club finds that 36 favor a proposal. A report claims about 65% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.

5. A survey of 50 people attending a poetry event finds that 33 favor a proposal. A report claims about 66% of all library members favor it. Evaluate the claim and suggest a better sampling plan.

6. A survey of 65 visitors to one sports field finds that 43 favor a proposal. A report claims about 66% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. A survey of 50 volunteers from its debate club finds that 33 favor a proposal. A report claims about 66% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.

8. A survey of 35 people attending a poetry event finds that 23 favor a proposal. A report claims about 66% of all library members favor it. Evaluate the claim and suggest a better sampling plan.

9. A survey of 55 visitors to one sports field finds that 36 favor a proposal. A report claims about 65% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. A survey of 35 volunteers from its debate club finds that 23 favor a proposal. A report claims about 66% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.

11. A survey of 35 people attending a poetry event finds that 23 favor a proposal. A report claims about 66% of all library members favor it. Evaluate the claim and suggest a better sampling plan.

12. A survey of 35 visitors to one sports field finds that 23 favor a proposal. A report claims about 66% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. A survey of 40 volunteers from its debate club finds that 26 favor a proposal. A report claims about 65% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.

14. A survey of 55 people attending a poetry event finds that 36 favor a proposal. A report claims about 65% of all library members favor it. Evaluate the claim and suggest a better sampling plan.

15. A survey of 60 visitors to one sports field finds that 40 favor a proposal. A report claims about 67% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. A survey of 55 volunteers from its debate club finds that 36 favor a proposal. A report claims about 65% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. A survey of 65 volunteers from its debate club finds that 43 favor a proposal. A report claims about 66% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.

2. A survey of 55 people attending a poetry event finds that 36 favor a proposal. A report claims about 65% of all library members favor it. Evaluate the claim and suggest a better sampling plan.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. A survey of 50 visitors to one sports field finds that 33 favor a proposal. A report claims about 66% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.

4. A survey of 40 volunteers from its debate club finds that 26 favor a proposal. A report claims about 65% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. A survey of 45 people attending a poetry event finds that 30 favor a proposal. A report claims about 67% of all library members favor it. Evaluate the claim and suggest a better sampling plan.

6. A survey of 35 visitors to one sports field finds that 23 favor a proposal. A report claims about 66% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.

Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. A survey of 55 volunteers from its debate club finds that 36 favor a proposal. A report claims about 65% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.

2. A survey of 35 people attending a poetry event finds that 23 favor a proposal. A report claims about 66% of all library members favor it. Evaluate the claim and suggest a better sampling plan.



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. A survey of 55 visitors to one sports field finds that 36 favor a proposal. A report claims about 65% of all residents of a town favor it. Evaluate the claim and suggest a better sampling plan.

4. A survey of 55 volunteers from its debate club finds that 36 favor a proposal. A report claims about 65% of all students at a school favor it. Evaluate the claim and suggest a better sampling plan.



Practice worked solutions

1. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.

2. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

3. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

4. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.



Practice worked solutions

5. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

6. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

7. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.

8. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

Practice worked solutions

9. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

10. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.

11. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

12. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

Practice worked solutions

13. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.

14. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

15. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

16. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.

Transfer worked solutions

1. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.

2. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

3. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

4. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.

Transfer worked solutions

5. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

6. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

Exit-ticket worked solutions

1. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.

2. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the membership list, plan for nonresponse, and state remaining uncertainty.

3. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random sample from an appropriate town-wide address frame, plan for nonresponse, and state remaining uncertainty.

4. The sample may not represent the target population.

Participation or location selects a particular group. The reported sample percentage does not justify the same population percentage. Use a random selection from the full enrollment list, plan for nonresponse, and state remaining uncertainty.



Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Evaluate a sampling method and the limits of a claim. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

The questions, contexts, explanations, and practice sequence are independently authored. No official test items, passages, answer choices, logos, or branded curriculum are reproduced. SAT is a College Board trademark; ACT is an ACT trademark. This resource is not affiliated with or endorsed by either organization.

Print US Letter at actual size. Separate student practice from the teacher keys. The PDF can be annotated in a private learning platform but does not contain fillable form fields. No paid software or device is required for print use.

Single-teacher classroom license: use copies with your own learners in a private classroom or LMS. Do not resell, share the complete file publicly, or redistribute it to other teachers. Suggested time and scoring guidance are instructional choices, not official exam timing, accommodations approval, or score prediction.

